RTMcompare

Ten surfaces, one master volume.

A surface-by-surface reference. Each row in this document is a real feature in the build you can install today — no marketing rounding-up, no "coming soon." If it's listed here, you can reach it from the app right now.

Use this when: you're trying to remember which panel does the per-band masking, or you're evaluating the tool against a spec, or you want to know what a feature is for in one line before you click into it.

SURFACE ONE

A/B Compare.

The flagship. Drop two files, level-matched. Every measurable difference between them, surfaced.

Level-matched playback
Both files normalised to −18 LUFS integrated before audition. Loud doesn't fake good.

LUFS-I, TP, LRA, MONO
The four delivery numbers, surfaced in the instrument row at the top of every screen.

Per-band masking
BS-RoFormer 4-stem separation (SDR 9.66) reveals where vocals, drums, bass, other lose energy. Per-band, not whole-mix.

Phase correlation over time
Full-track plot plus per-band phase ribbon. Catches phase issues scalar correlation hides.

Vectorscope
XY mid/side scope with peak-hold. Width and centre-energy at a glance.

Streaming-normalisation preview
Spotify, Apple, Amazon, Tidal, YouTube. Hear exactly how each platform will play your master.

Inter-sample peak meter
4× oversampled true-peak detection (BS.1770-4). The Apply-and-bounce limiter steps to 16× polyphase Kaiser internally for sub-0.05 dB ceiling accuracy on the rendered master.

AAC encode preview
Render through Apple's AAC encoder, A/B against the source.

Engineer-profile matching
Serban Ghenea, Chris Lord-Alge, your own (built with RTMprofile). Match-score plus concrete EQ moves.

EQ-move export
FabFilter Pro-Q text, CSV, JSON. Or apply-and-bounce a corrected master WAV in one click.

SURFACE TWO

Single-File QC.

Drop one file. Get a deep clinical pass. For when there is no reference, only the file and a deadline.

Click + glitch timeline
FLOW v2 LPC-residual detector. Peaks plotted with click-to-transport jump. Drum-friendly — no more snare false positives.

Distortion detection
Clipping, ISR, harmonic. Severity rating. Frequency band where it sits.

Mains hum + harmonics
50 / 60 Hz fundamental plus 3rd, 5th, 7th. Catches grounding problems you never thought to listen for.

Transfer-artefact detection
Wow, flutter, DC drift, tape transport, print-through. For analog-sourced masters.

Generation-loss detection
Prior AAC or MP3 encoding fingerprints. Surfaces lossy ancestors.

Key, BPM, harmonic ladder
For sync pitching and metadata.

Mono-compat waterfall
Per band, not just a scalar. See exactly where the mono fold collapses.

Stereo image + phase bands
Image width and per-band phase, side by side.

SURFACE THREE & FOUR

Album Batch + Cohort.

A folder in. A sortable table out. Drift detection across the album.

Sortable overview table
LUFS, true peak, LRA, ISRC, duration, sample rate, bit depth, outlier flags.

One rotating song tab
← → to step through. Lazy deep analysis cached across rotations.

Per-song + album notes
Embedded in every PDF export. Travels with the file.

.rtmalbum.json sessions
Save and load every analysis, every note, the A/B ref, cohort ref, DMR state. Re-open in a year and pick up exactly where you stopped.

Cohort heatmap
Per-track distance across 31 bands. Sort by drift to find the outliers.

RMS distance column
A single ranked metric for class-wide consistency.

SURFACE FIVE & SIX

Delivery Manifest + Atmos.

The boring stuff that saves you the midnight Apple rejection. Plus the immersive corner for ADM BWF work.

Three-way diff
Audio-embedded metadata ⬄ distributor manifest ⬄ batch-internal ISRC set.

Title-casing drift
Feat. vs feat. is enough for Apple to auto-cancel. Surfaced as a blocker.

ISRC collisions + reuse
Across the album AND across prior releases (history at ~/.rtm/isrc-history.json).

Duration mismatches
Between audio file and manifest.

Missing rows on either side
Surfaces orphans audio-side or manifest-side.

P-line / C-line check
Copyright string drift.

Ship-Ready PDF + Corrected CSV
Attach to the delivery ticket. The distributor can re-ingest the CSV.

ADM BWF parsing
Bed, objects, trajectories, channel mapping.

Binaural-headroom estimate
Early-warning ILD downmix (no HRTF). Apple's Atmos guideline is < −1 dBTP on their renderer's binaural deliverable; this is a fast sanity-check, not a substitute.

Atmos Preflight hard-checks
Object count □ 118, LFE routing, bed layout, SR = 48 kHz, BD □ 24.

Per-object anomaly detection
Hot, silent, static, dark objects. Usually mix mistakes, occasionally artistic intent.

Verdict surfaces and visual chrome.

How RTMcompare tells you what to change, plays files back, and gates delivery.

Engineer-target curves
Match-score against the chosen engineer profile.

Concrete EQ moves
Frequency, gain, Q. Exportable.

Apply-and-bounce
One click. Corrected WAV in your render folder.

A/B Player everywhere
Same engine, same shortcuts. B tracks active context. A is whatever you picked.

Live TP meter
Instantaneous and 2-second peak-hold on the transport.

Triage Mode (optional)
Ready-to-Deliver verdict, Attention list, per-DSP spec profile.

Header palette + shortcuts (5.5)
Reference dropdown · "+ New analysis" · Search palette (□K) · keyboard-shortcut sheet (?) — all in the header. Sticky Advanced QC across sessions.

EQ Preview level-match pill
A/B the EQ tweak with and without level-matching in one click. The pill is on the panel, not buried in the gear-icon menu.

Console Didone Shell
Two-row header, Didone instrument metrics, cover-state empty screen, single gold per screen.

v1 Classic shell
localStorage[rtm-shell] = v1 returns the v5.1.x markup byte-for-byte if you want it back.

RTMprofile companion
Standalone app. Builds your engineer profile from 5+ finished masters.

RTMsend companion
VST3 / AU plugin. One-button bridge from your DAW into RTMcompare. ARA-aware.

COLOPHON

What this is built on.

The stack underneath, named explicitly because the credit matters.

```
Electron · React 19 · TypeScript · Tailwind v4 · Vite
Python 3.11 · NumPy · SciPy · librosa · pyloudnorm
BS-RoFormer 4-stem (audio-separator) · Demucs (fallback)
JUCE 7 (RTMsend) · electron-builder · electron-rebuild
```

Wordmark and hero numerals set in Instrument Serif. Body and labels in Instrument Sans. Tabular data in Geist Mono. The single antique-gold accent is reserved for one element per screen — the active delivery-target chip, the verdict-violation flag, or the icon mark. Never two at once.